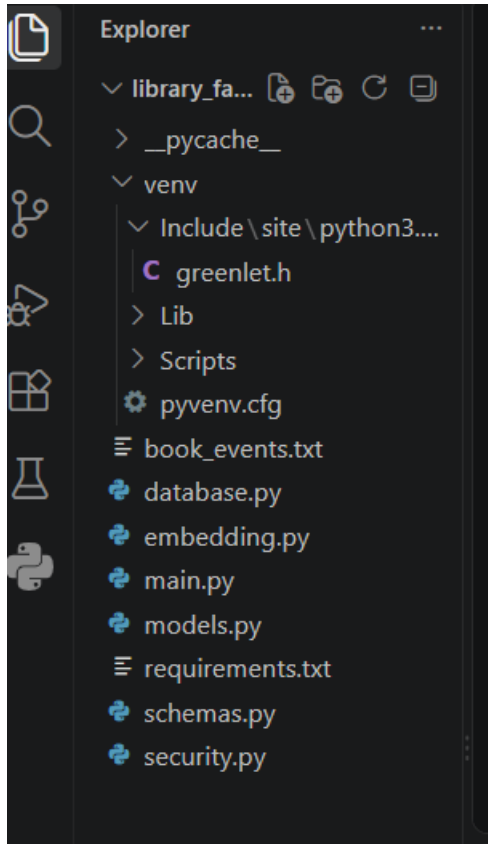


PRACTICAL WEEK-5

1. Project Structure



2. Installed Packages

```
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\starlette\routing.py", line 648, in lifespan
    async with self.lifespan_context(app) as maybe_state:
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 265, in __aenter__
    await self.router.startup()
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\fastapi\routing.py", line 6375, in startup
    await handler()
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\main.py", line 38, in startup
    async with engine.begin() as connection:
File "C:\Users\madhulika\AppData\Local\Programs\Python\Python311\Lib\contextlib.py", line 210, in __aenter__
    return await anext(self.gen)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\engine.py", line 1068, in begin
    async with conn:
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\ext\asyncio\base.py", line 121, in __aenter__
    return await self.start(is_ctxmanager=True)
```

```
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 144, in __init__
    self._dbapi_connection = engine.raw_connection()
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\engine\base.py", line 3319, in raw_connection
    return self.pool.connect()
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 448, in connect
    return _ConnectionFairy.checkout(self)
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 1272, in _checkout
    fairy = _ConnectionRecord.checkout(pool)
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\sqlalchemy\pool\base.py", line 712, in checkout
    rec = pool._do_get()
    ~~~~~
```

```

File "asyncmy/connection.py", line 976, in connect
File "asyncmy/connection.py", line 954, in asyncmy.connection.Connection.connect
File "asyncmy/connection.py", line 1316, in _request_authentication
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\asyncmy\auth.py", line 246, in caching_sha2_password_auth
    pkt = await _roundtrip(conn, data)
    ~~~~~
File "C:\Users\madhulika\OneDrive\Desktop\DBBS\library_fastapi_project\venv\Lib\site-packages\asyncmy\auth.py", line 116, in _roundtrip
    pkt = await conn.read_packet()
    ~~~~~
File "asyncmy/connection.py", line 1087, in read_packet
File "asyncmy/protocol.py", line 956, in asyncmy.protocol.MysqlPacket.raise_for_error
File "asyncmy/protocol.py", line 957, in asyncmy.protocol.MysqlPacket.raise_for_error
File "asyncmy/errors.py", line 141, in asyncmy.errors.raise_mysql_exception
File "asyncmy/errors.py", line 156, in asyncmy.errors.raise_mysql_exception
sqlalchemy.exc.OperationalError: (asyncmy.errors.OperationalError) (1045, "Access denied for user 'root'@'localhost' (using password: YES)")

```

3. MySQL Setup

FPS 0 workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK-1 WEEK-2 SQL File 4* WEEK-5* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown

```

1 • CREATE DATABASE library_db;
2 • USE library_db;
3 • CREATE TABLE members (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     username VARCHAR(100) UNIQUE NOT NULL,
6     password_hash VARCHAR(255) NOT NULL,
7     role VARCHAR(30) DEFAULT 'member'
8 );

```

Limit to 1000 rows

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```

8 );
9 • CREATE TABLE books (
10     id INT AUTO_INCREMENT PRIMARY KEY,
11     title VARCHAR(255) NOT NULL,
12     author VARCHAR(255) NOT NULL,
13     isbn VARCHAR(13) UNIQUE NOT NULL,
14     price DECIMAL(10,2) NOT NULL,
15     published_date DATE NOT NULL,
16     description TEXT,
17     embedding JSON
18 );
19 • SHOW DATABASES;

```

SQLAdditions

Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.

Result Grid

Database

- bookflow_db
- bookflow_dbh
- information_schema
- kludb
- library_db
- madhulika_2520090056
- mysql
- performance_schema

Result 1 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	20:05:05	USE library_db	0 row(s) affected	0.000 sec
3	20:05:19	CREATE TABLE members (id INT AUTO_INCREMENT PRI...	0 row(s) affected	0.141 sec
4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected	0.062 sec
5	20:05:55	SHOW DATABASES	10 row(s) returned	0.000 sec / 0.000 sec

The screenshot shows a database management tool interface with a sidebar on the left containing sections: MANAGEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), INSTANCE (Startup / Shutdown, Server Logs, Options File), PERFORMANCE (Dashboard, Performance Reports, Performance Schema Setup), Administration (Schemas), and Information (No object selected). The main window displays a SQL script in a text editor with line numbers 10 to 21. The script defines a table 'books' and then executes 'SHOW DATABASES;', 'USE library_db;', and 'SHOW TABLES;'. Below the editor, a 'Result Grid' shows the output of the 'SHOW TABLES;' command, listing 'books' and 'members' under the 'Tables_in_library_db' column. At the bottom, an 'Output' pane shows a log of actions: creating the 'books' table, showing databases, using 'library_db', and showing tables, with corresponding row counts.

```

10      id INT AUTO_INCREMENT PRIMARY KEY,
11      title VARCHAR(255) NOT NULL,
12      author VARCHAR(255) NOT NULL,
13      isbn VARCHAR(13) UNIQUE NOT NULL,
14      price DECIMAL(10,2) NOT NULL,
15      published_date DATE NOT NULL,
16      description TEXT,
17      embedding JSON
18  );
19  • SHOW DATABASES;
20  • USE library_db;
21  • SHOW TABLES;

```

Tables_in_library_db
books
members

#	Time	Action	Message
✓ 4	20:05:32	CREATE TABLE books (id INT AUTO_INCREMENT PRIM...	0 row(s) affected
✓ 5	20:05:55	SHOW DATABASES	10 row(s) returned
✓ 6	20:07:46	USE library_db	0 row(s) affected
✓ 7	20:07:52	SHOW TABLES	2 row(s) returned

4. Database.py

The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'library_fastapi_project' with various files and folders. The code editor displays the contents of 'database.py', which imports 'AsyncSession' and 'DeclarativeBase' from 'sqlalchemy.ext.asyncio' and 'sqlalchemy.orm' respectively. It defines a 'DATABASE_URL' constant, creates an 'engine' using 'create_async_engine', and defines an 'AsyncSessionLocal' using 'async_sessionmaker'. A 'Base' class is defined as a subclass of 'DeclarativeBase'. An asynchronous function 'get_db()' is defined to return a session from 'AsyncSessionLocal'.

```

1  from sqlalchemy.ext.asyncio import AsyncSession, async_sessionmaker, create_async_engine
2  from sqlalchemy.orm import DeclarativeBase
3
4  DATABASE_URL = (
5      "mysql+asyncmy://root:Madhu%40248"
6      "@localhost:3306/library_db"
7  )
8
9  engine = create_async_engine(
10     DATABASE_URL,
11     echo=True
12 )
13
14  AsyncSessionLocal = async_sessionmaker(
15     engine,
16     expire_on_commit=False
17 )
18
19
20  class Base(DeclarativeBase):
21     pass
22
23
24  async def get_db():
25     async with AsyncSessionLocal() as session:
26         yield session

```

5. Embedding.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

embedding.py > text_to_vector
1  import hashlib
2  import math
3
4
5  VECTOR_DIM = 384
6
7
8  def text_to_vector(text: str):
9      vector = []
10
11      for i in range(VECTOR_DIM):
12          h = hashlib.sha256(
13              f"{text}-{i}".encode()
14          ).digest()
15
16          value = int.from_bytes(h[:4], "big") / (2**32 - 1)
17          vector.append(value)
18
19      norm = math.sqrt(
20          sum(x * x for x in vector)
21      )
22
23      if norm == 0:
24          return vector
25
26      return [x / norm for x in vector]
```

6. Main.py

```
library_fa...
├── __pycache__
├── venv
├── Include\site\python3....
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
├── security.py
└── ...

main.py > login
1  from fastapi import (
2      BackgroundTasks,
3      Depends,
4      FastAPI,
5      HTTPException,
6      Query,
7      status
8  )
9
10 from fastapi.security import OAuth2PasswordRequestForm
11
12 from sqlalchemy import select
13 from sqlalchemy.ext.asyncio import AsyncSession
14
15 from database import Base, engine, get_db
16 from embedding import text_to_vector
17 from models import Book, Member
18 from schemas import BookSchema, MemberRegister, TokenResponse
19
20 from security import (
21     create_access_token,
22     get_current_member,
23     hash_password,
24     verify_password
25 )
26
27 import math
28
29
30 app = FastAPI(
31     title="Library Backend & Security API",
32     version="1.0"
33 )
34
35
36 @app.on_event("startup")
```

7. Models.py

```
library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

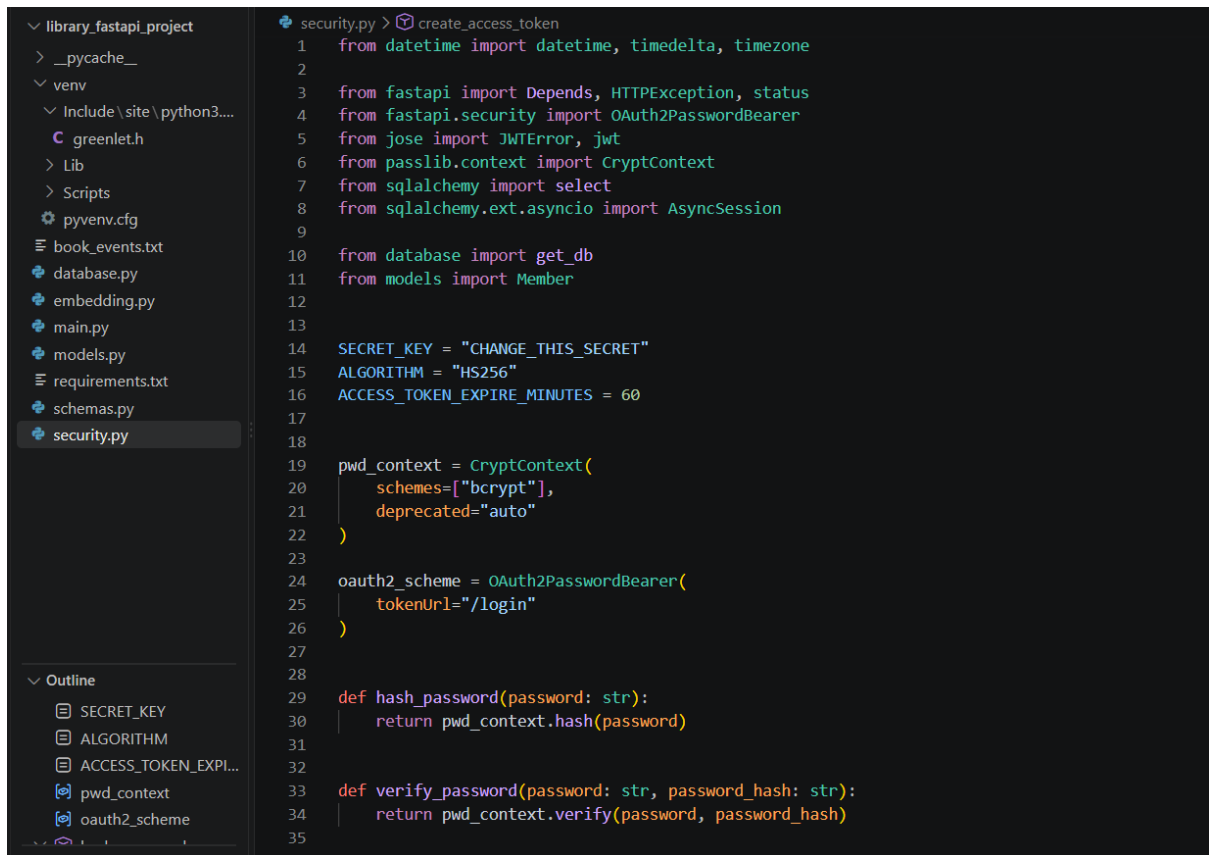
```
models.py > Book
1  from sqlalchemy import Column, Integer, String, Text, DECIMAL, Date, JSON
2  from database import Base
3
4
5  class Member(Base):
6      __tablename__ = "members"
7
8      id = Column(Integer, primary_key=True, index=True)
9      username = Column(String(100), unique=True, nullable=False)
10     password_hash = Column(String(255), nullable=False)
11     role = Column(String(30), default="member")
12
13
14     class Book(Base):
15         __tablename__ = "books"
16
17         id = Column(Integer, primary_key=True, index=True)
18         title = Column(String(255), nullable=False)
19         author = Column(String(255), nullable=False)
20         isbn = Column(String(13), unique=True, nullable=False)
21         price = Column(DECIMAL(10, 2), nullable=False)
22         published_date = Column(Date, nullable=False)
23         description = Column(Text)
24         embedding = Column(JSON)
```

8. Schemas.py

```
Explorer
├── library_fastapi_project
├── __pycache__
├── venv
├── Include\site\python3...
├── greenlet.h
├── Lib
├── Scripts
├── pyvenv.cfg
├── book_events.txt
├── database.py
├── embedding.py
├── main.py
├── models.py
├── requirements.txt
├── schemas.py
└── security.py
```

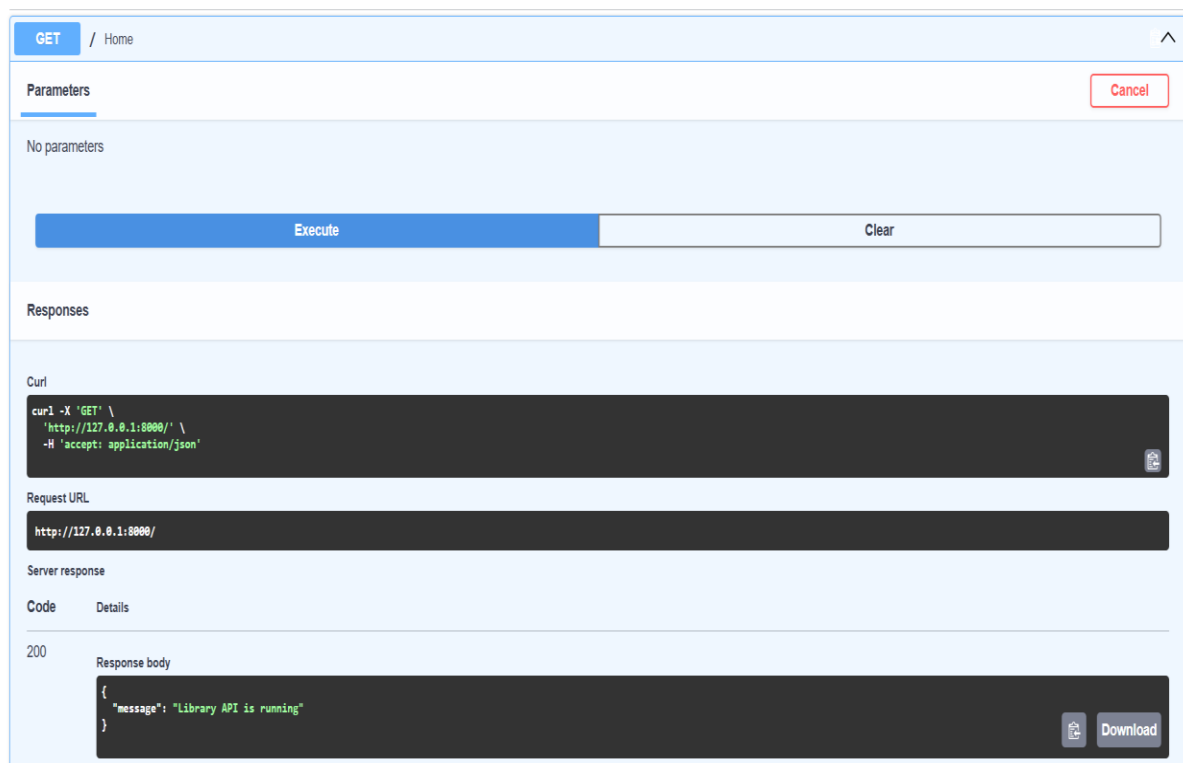
```
schemas.py > MemberRegister
1  from pydantic import BaseModel, Field, field_validator
2  from datetime import date
3
4
5  class BookSchema(BaseModel):
6      title: str
7      author: str
8      isbn: str = Field(min_length=13, max_length=13)
9      price: float = Field(gt=0)
10     published_date: date
11     description: str | None = None
12
13     @field_validator("published_date")
14     @classmethod
15     def validate_published_date(cls, value):
16         if value > date.today():
17             raise ValueError("published_date cannot be in the future")
18         return value
19
20
21     class MemberRegister(BaseModel):
22         username: str
23         password: str = Field(min_length=6)
24
25
26     class TokenResponse(BaseModel):
27         access_token: str
28         token_type: str
```

9. Security.py



```
security.py > create_access_token
1  from datetime import datetime, timedelta, timezone
2
3  from fastapi import Depends, HTTPException, status
4  from fastapi.security import OAuth2PasswordBearer
5  from jose import JWTError, jwt
6  from passlib.context import CryptContext
7  from sqlalchemy import select
8  from sqlalchemy.ext.asyncio import AsyncSession
9
10 from database import get_db
11 from models import Member
12
13
14 SECRET_KEY = "CHANGE_THIS_SECRET"
15 ALGORITHM = "HS256"
16 ACCESS_TOKEN_EXPIRE_MINUTES = 60
17
18
19 pwd_context = CryptContext(
20     schemes=["bcrypt"],
21     deprecated="auto"
22 )
23
24 oauth2_scheme = OAuth2PasswordBearer(
25     tokenUrl="/login"
26 )
27
28
29 def hash_password(password: str):
30     return pwd_context.hash(password)
31
32
33 def verify_password(password: str, password_hash: str):
34     return pwd_context.verify(password, password_hash)
35
36
```

10. Test Registration



POST/registerRegister

Parameters

CancelReset

No parameters

Request bodyrequired

application/json

Edit ValueSchema

```
{
  "username": "Madhu",
  "password": "library123"
}
```

Execute

Clear

Responses

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8080/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "Madhu",
    "password": "library123"
  }'
```

Request URL

http://127.0.0.1:8080/register

Server response

CodeDetails

200

Response body

```
{
  "message": "Member registered successfully",
  "username": "Madhu",
  "role": "member"
}
```

Download

Response headers

```
content-length: 79
content-type: application/json
date: Mon, 14 Sep 2026 16:05:16 GMT
server: uvicorn
```

Responses

11. Test Login

POST/login Login

Parameters

CancelReset

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: ^password\$

password

☐ Send empty value

username * required

string

Madhu

password * required

string(\$password)

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

Madhu

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=Madhu&password=library123&scope=&client_id=Madhu&client_secret=library123'
```

Request URL

http://127.0.0.1:8000/login

Server response

Code

Details

200

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IWRXVjQ9LmV7ZmI0IjNYMmR5dSI6InR5bGU0Ij0iZW1iZXI1LCJleHAiOjE3ODk0MDU5Nzh9,BSyLTpIyOmAoCo0hwV3PUg1xVb9I2uZBd08fy3Vx4",
  "token_type": "bearer"
}
```

Download

Response headers

```
content-length: 186
content-type: application/json
date: Mon, 14 Sep 2026 16:12:57 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls Accept header

Example Value

Schema

12. Test POST/Books

POST /books Create Book

Parameters

No parameters

Request body required

application/json

Edit Value Schema

```
{
  "title": "Python for Beginners",
  "author": "John Smith",
  "isbn": "123",
  "price": -100.00,
  "published_date": "2024-06-10",
  "description": "An introduction to Python programming."
}
```

Execute

Clear

Responses

```
curl -X POST \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJWYWRodSI6InJvbGUiOiI7ZWliZXIiOiJleHA1OjE3MDk0NDYwMjZ9.Lq7z0U6THNkMxQh8zXEqTX64tTGk1sHhu4JgWnB1I0' \
  -H 'Content-Type: application/json' \
  -d '{
    "title": "Python for Beginners",
    "author": "John Smith",
    "isbn": "123",
    "price": 450.00,
    "published_date": "2024-06-10",
    "description": "An introduction to Python programming."
  }'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

422

Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "string_too_short",
      "loc": [
        "body",
        "isbn"
      ],
      "msg": "String should have at least 13 characters",
      "input": "123",
      "ctx": {
        "min_length": 13
      }
    }
  ]
}
```

Download

Response headers

```
content-length: 150
content-type: application/json
date: Mon, 14 Sep 2026 16:28:41 GMT
server: uvicorn
```

GET

/books/search

Search Books

Parameters

Cancel

Name	Description
q * required	
string	q = python
(query)	minLength: 1
limit	
integer	5
(query)	maximum: 20 minimum: 1

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/books/search?q=q%20python&limit=5' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/books/search?q=q%20python&limit=5
```

Server response

13. book_event.txt

library_fastapi_project

> __pycache__

venv

Include\site\python3....

greenlet.h

Lib

Scripts

pyvenv.cfg

book_events.txt

database.py

embedding.py

main.py

models.py

requirements.txt

schemas.py

security.py

book_events.txt

1 New Book Added: id=1, title=Python for Beginners

2

